Engineering Challenges

Section Européenne – Sciences de l'Ingénieur

EC01 to EC30

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EC01 – Renewable Energy

Can solar power meet tomorrow's energy needs?

Learning Objectives

- Explain how a photovoltaic cell and a grid-connected PV system operate.
- Interpret capacity data and calculate annual additions.
- Discuss intermittency, storage, grid integration and recycling.

Engineering News

Global renewable capacity grew by a record 692 GW in 2025. Solar power alone accounted for about 510 GW of additions, nearly three-quarters of the renewable total. The engineering challenge is now not only to manufacture more panels, but also to connect variable generation to stronger and more flexible grids.

Main Article

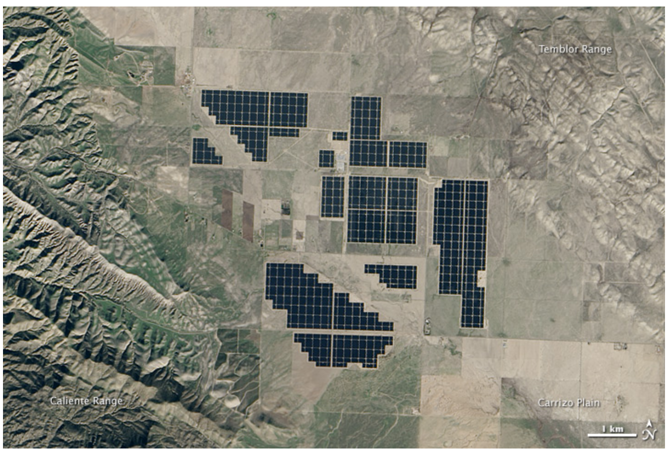
A photovoltaic (PV) cell converts light directly into electricity. Most commercial cells are manufactured from silicon, a semiconductor. When photons provide enough energy, electrons can move through the material. The internal electric field of the p–n junction separates the electrical charges and creates a direct current.

A PV module connects many cells together. Modules are combined into strings and arrays to reach the required voltage and power. However, a module rarely operates at a fixed point. Its voltage-current characteristic changes with solar irradiance and temperature. A maximum power point tracker continuously adjusts the operating point so that the array delivers as much power as possible.

The generated electricity is direct current, whereas buildings and public grids use alternating current. An inverter therefore converts DC into AC, synchronises the voltage and frequency with the grid, and provides protection and monitoring functions. A battery may store surplus electricity, but it increases cost, material use and conversion losses.

Solar power is variable: output changes with clouds, season and time of day. Engineers integrate large amounts of PV using forecasting, flexible demand, interconnections, storage and controllable generation. The objective is not to make every installation independent, but to balance the complete electricity system.

The environmental impact of a PV system must be assessed over its entire life cycle. Manufacturing requires energy and materials, while end-of-life modules contain glass, aluminium, copper, polymers and semiconductor materials. Designing modules that are efficient, durable, repairable and recyclable is therefore a major engineering goal.



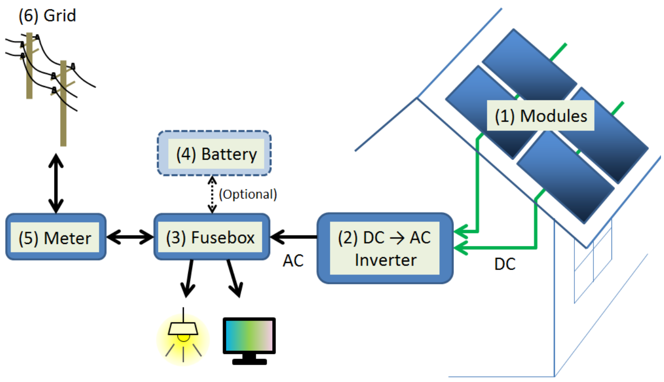
Topaz Solar Farm, California. NASA image, public domain.

Engineering Focus

Designing a photovoltaic installation
Engineers must optimise several parameters simultaneously:

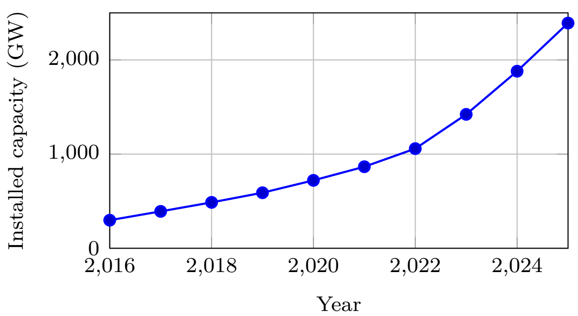
- panel orientation and tilt;
- inverter sizing and MPPT architecture;
- cable cross-sections and electrical losses;
- thermal behaviour and ventilation;
- protection and isolation devices;
- monitoring and maintenance strategy;
- expected lifetime and recyclability.

A good design minimises the levelised cost of electricity while ensuring safety, reliability and high annual energy yield.



Simplified residential PV system. S-kei, CC0.

Global solar PV capacity



Source: IRENA, Renewable Capacity Statistics 2026.

Did You Know?

The solar energy reaching the Earth in roughly one hour is comparable to humanity's annual energy use. The main engineering challenge is therefore to capture, convert, store and distribute only a small fraction of this resource efficiently.

Key Figures

Commercial module efficiency	20–24%
Typical module lifetime	25–30 years
Typical inverter efficiency	97–99%
Annual module degradation	0.3–0.5%/year
Global PV capacity in 2025	≈ 2.4 TW

Useful Vocabulary

solar cell	cellule photovoltaïque
irradiance	irradiance
inverter	onduleur
grid	réseau électrique
storage	stockage
yield	production énergétique
lifetime	durée de vie
recycling	recyclage

Questions

1. Explain the role of the p–n junction.
2. Why does a PV installation need an inverter?
3. What is the purpose of MPPT control?
4. Use the graph to estimate the increase between 2020 and 2025.
5. Give three technical solutions for integrating variable solar power.
6. Why must environmental assessment include manufacturing and end-of-life?

Engineering Challenge

Your school consumes 410 MWh of electricity each year and has 850 m^2 of usable roof area. Prepare a preliminary PV study:

1. estimate the peak power that could be installed;
2. estimate annual energy production;
3. discuss the likely self-consumption ratio;
4. decide whether battery storage is justified;
5. propose useful monitoring indicators.

Present your conclusions as a short engineering report.

Speaking Task

Working in pairs, prepare a five-minute presentation answering the question: "Can Europe rely mainly on solar electricity by 2050?" Support your arguments with technical evidence, environmental impacts, economic considerations and examples from different European countries.

EC02 – Hydrogen

Can hydrogen decarbonise heavy transport?

Learning Objectives

- Distinguish grey, blue and green hydrogen.
- Explain the operating principles of electrolysers and fuel cells.
- Compare hydrogen and battery-electric energy chains.

Engineering News

Low-emission hydrogen projects are being developed for fertiliser production, steel, shipping and heavy road transport. Their success depends not only on electrolyser capacity, but also on electricity supply, storage, transport and final demand.

Main Article

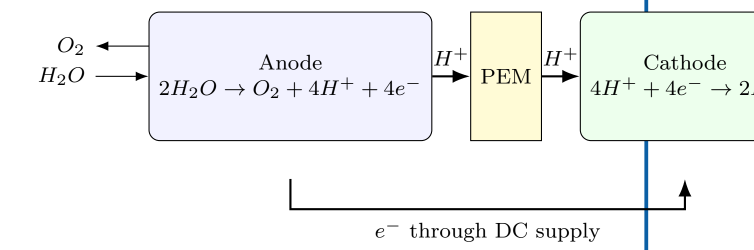
Hydrogen is an **energy carrier**, not a primary source of energy. It must be manufactured from another resource before it can be stored and used. Most hydrogen is still produced from natural gas by steam methane reforming. If the resulting carbon dioxide is released, the product is called grey hydrogen. Capturing part of the CO_2 leads to so-called blue hydrogen. Green hydrogen is produced by electrolysis powered by low-carbon renewable electricity.

In a proton-exchange-membrane (PEM) electrolyser, water enters the anode side. An electric current separates the water molecules into oxygen, protons and electrons. The membrane allows protons to cross while forcing electrons through an external circuit. Hydrogen is formed at the cathode.

A PEM fuel cell performs the reverse overall reaction. Hydrogen is supplied to the anode and oxygen from air reaches the cathode. The electrochemical reaction produces electricity, water and heat without combustion. Several cells are assembled in a stack to obtain the required voltage and power.

Hydrogen has a high specific energy by mass and enables rapid refuelling. However, its low volumetric density requires compression, liquefaction or conversion into another carrier. Each conversion consumes energy. Consequently, a battery-electric vehicle is usually more efficient from electricity source to wheels, while hydrogen can remain attractive when long range, high payload and short refuelling stops are essential.

PEM electrolyser



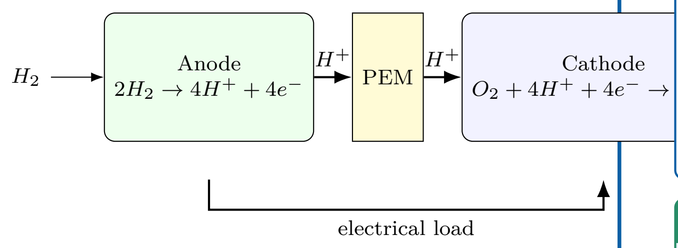
Engineering Focus

The complete hydrogen value chain

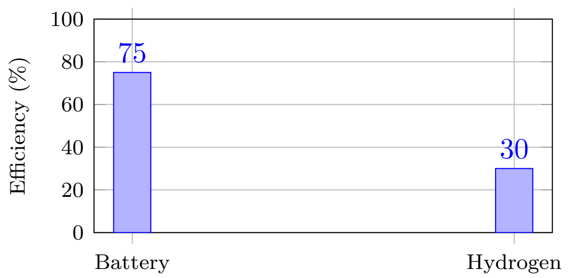
1. Generate low-carbon electricity.
2. Produce hydrogen by electrolysis.
3. Purify and compress the gas.
4. Store and distribute it safely.
5. Convert it into useful energy in a fuel cell or industrial process.

Engineers optimise efficiency, safety, materials, cost and infrastructure at every stage. Hydrogen leaks must also be controlled because the molecule is very small and has a wide flammability range in air.

PEM fuel cell



Electricity-to-wheel efficiency



Illustrative whole-chain values; actual performance depends on equipment and operating conditions.

Did You Know?

Hydrogen contains about 120 MJ of energy per kilogram on a lower-heating-value basis, but storing one kilogram requires a much larger tank than storing an equivalent amount of liquid fuel.

Key Figures

Hydrogen LHV	≈ 120 MJ/kg
Electrolyser efficiency	60–75%
Fuel-cell electrical efficiency	40–60%
Typical vehicle storage pressure	350 or 700 bar

Useful Vocabulary

electrolyser	électrolyseur
fuel cell	pile à combustible
stack	empilement de cellules
compressed gas	gaz comprimé
refuelling	ravitaillement
leak detection	détection de fuite

Questions

1. Why is hydrogen described as an energy carrier?
2. Compare grey, blue and green hydrogen.
3. Explain the role of the PEM in an electrolyser.
4. What products leave a fuel cell?
5. Why is the battery chain usually more efficient?
6. In which transport applications can hydrogen remain attractive?

Engineering Challenge

A logistics company operates 120 long-haul trucks. Compare battery-electric and hydrogen fuel-cell solutions using efficiency, range, payload, refuelling time, infrastructure and life-cycle emissions. Present a justified engineering recommendation.

Speaking Task

Debate: "Hydrogen will replace batteries for heavy transport." Each team must use at least three technical arguments and respond to one opposing argument.

Sources and further reading

IEA — Global Hydrogen Review: online resource

;

U.S. Department of Energy — Hydrogen and Fuel Cell Technologies Office: online resource

;

European Commission — Hydrogen: online resource

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EC03 – Wind Energy

How do engineers harvest the wind?

Learning Objectives

- Explain how aerodynamic lift produces rotor torque.
- Identify the main subsystems of a modern wind turbine.
- Interpret a wind-turbine power curve.
- Compare onshore, fixed-bottom offshore and floating offshore solutions.

Engineering News

Modern offshore wind turbines now reach ratings above 15 MW. Floating foundations make it possible to exploit deep-water sites where winds are stronger and more regular, but they also introduce new challenges in mooring, maintenance and grid connection.

Key Figures

Rotor diameter	120–250 m
Rated power	3–18 MW
Cut-in speed	3–4 m/s
Rated speed	11–15 m/s
Cut-out speed	20–25 m/s
Capacity factor	30–60%

Main Article

A wind turbine converts the kinetic energy of moving air into electrical energy. The blades are shaped like aerofoils. As air flows around a blade, a pressure difference creates lift. The tangential component of this force produces a torque on the rotor.

The available power in the wind is

$$P_{\text{wind}} = \frac{1}{2}\rho A v^3,$$

where ρ is air density, A the swept area and v wind speed. The cubic term explains why site selection is critical: a small increase in average wind speed can produce a large increase in annual energy output.

The rotor drives either a gearbox and a high-speed generator or a direct-drive generator. Power electronics convert the variable-frequency output into electricity compatible with the grid. Pitch control adjusts the blade angle, while yaw control rotates the nacelle so that the rotor faces the wind.

Engineering Focus

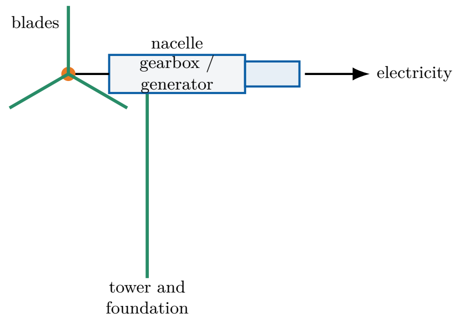
Main subsystems

1. Rotor, blades and hub;
2. Main shaft and bearings;
3. Gearbox or direct-drive generator;
4. Converter and transformer;
5. Pitch and yaw actuators;
6. Tower, foundations and monitoring system.

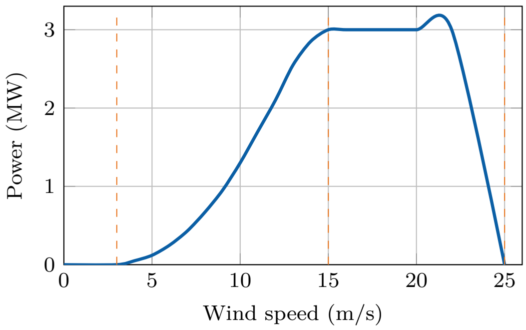
Did You Know?

No turbine can capture all the power in the wind. The theoretical maximum is the **Betz limit**: 59.3% of the kinetic power crossing the rotor area.

Wind-turbine architecture



Typical power curve



Cut-in, rated and cut-out regions determine how the turbine operates safely.

Useful Vocabulary

blade	pale
hub	moyeu
gearbox	multiplicateur
yaw	orientation de la nacelle
pitch	calage des pales
swept area	surface balayée
capacity factor	facteur de charge

Questions

1. Why does wind power depend on the cube of wind speed?
2. What is the purpose of pitch control?
3. What is the purpose of yaw control?
4. Interpret the three regions of the power curve.
5. Compare onshore and offshore wind farms.

Engineering Challenge

A coastal region plans a 500 MW wind farm. Compare an onshore solution, a fixed-bottom offshore solution and a floating offshore solution. Consider energy yield, foundations, maintenance, grid connection, environmental impact and public acceptance. Present a justified recommendation.

Speaking Task

Debate the statement: “Large wind farms should be built mainly offshore.” Use technical, economic and environmental arguments.

Sources

Global Wind Energy Council: online resource

IEA Wind: online resource

EC04 – Nuclear Energy

Can nuclear power support a low-carbon future?

Learning Objectives

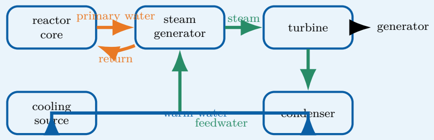
- Explain nuclear fission and the controlled chain reaction.
- Identify the energy and fluid flows in a pressurised water reactor (PWR).
- Interpret lifecycle-emission data and discuss the role of nuclear power.

Engineering News

Several countries are studying new reactor programmes, lifetime extensions and Small Modular Reactors (SMRs). The central engineering questions concern construction time, financing, safety demonstration, fuel supply, waste management and integration with an electricity system containing more variable renewable generation.

Engineering Focus

Energy conversion in a PWR



The primary circuit transports nuclear heat, the secondary circuit produces steam and the cooling circuit rejects unused heat. The reactor is therefore both a nuclear system and a large thermal power plant.

Did You Know?

Stopping fission does not immediately stop heat production. Residual heat falls rapidly, but cooling remains essential for hours, days and longer after shutdown.

Main Article

A nuclear power station converts the energy released by **nuclear fission** into electricity. When a uranium-235 nucleus absorbs a neutron, it can split into two lighter nuclei, release energy and emit additional neutrons. These neutrons may trigger further fissions. In a reactor, the chain reaction is kept stable rather than allowed to grow uncontrolled.

In a pressurised water reactor, fuel assemblies are placed inside the reactor vessel. Water in the primary circuit removes heat from the core. It is maintained at high pressure so that it remains liquid at a temperature of roughly 300 °C. In the steam generator, this primary water transfers heat to a separate secondary circuit. The secondary water boils and produces steam, which drives a turbine connected to an electrical generator.

After the turbine, a condenser transforms the steam back into liquid water. A third cooling circuit carries waste heat towards a river, the sea or a cooling tower. The three circuits have different functions and are kept physically separate.

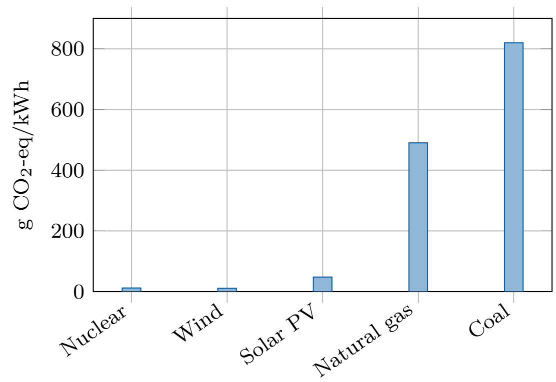
Reactor power is controlled by adjusting neutron absorption. Control rods contain materials that absorb neutrons, while soluble boron may also be used in the primary water. Even after the chain reaction stops, radioactive decay continues to release **residual heat**; cooling must therefore remain available.

Safety relies on several independent barriers and systems: the fuel matrix, the metal cladding, the primary circuit boundary and the containment building. Engineers also analyse extreme events, redundancy, component ageing, human factors, cybersecurity and long-term waste management.

Key Figures

Typical French PWR output	900–1450 MW
Primary-water temperature	about 300 °C
Primary pressure	about 155 bar
Typical capacity factor	70–90%
Design operating life	several decades

Indicative lifecycle emissions



Indicative median lifecycle values based on IPCC assessments. Values vary with technology, location and methodology.

Fission and fusion

Fission splits heavy nuclei and is used in present commercial reactors. **Fusion** combines light nuclei and powers the Sun. Fusion research aims to produce controlled net energy, but it is not yet a commercial electricity source.

Useful Vocabulary

chain reaction	réaction en chaîne
reactor vessel	cuve du réacteur
fuel assembly	assemblage combustible
control rod	barre de contrôle
steam generator	générateur de vapeur
containment	enceinte de confinement
residual heat	puissance résiduelle
spent fuel	combustible usé

Questions

1. What happens during nuclear fission?
2. Why must the primary water remain under high pressure?
3. Explain why the PWR contains several separate water circuits.
4. What is the role of control rods?
5. Why is cooling required after reactor shutdown?
6. Use the graph to compare nuclear power with fossil generation.
7. Give two advantages and two limitations of SMRs.

Engineering Challenge

A region must replace 8 GW of fossil-fuelled generation while reducing lifecycle CO_2 emissions and maintaining electricity supply during winter evenings. Propose a balanced mix using nuclear, wind, solar, storage, interconnections and demand response. Justify:

1. installed capacity;
2. expected availability;
3. flexibility and reserve requirements;
4. environmental and social constraints;
5. construction sequence and major risks.

Present the result as a two-page engineering recommendation.

Speaking Task

Debate: "Europe should build new nuclear power stations." One team supports the proposal and one team opposes it. Use evidence about safety, climate, cost, waste, construction time and security of supply.

Sources and further reading

IAEA - Nuclear Power Reactors: online resource

OECD Nuclear Energy Agency: online resource

IPCC - Energy Systems: online resource

EC05 – Energy Storage

Why storing energy is key to the energy transition

Learning Objectives

- Explain why electrical grids need energy storage.
- Compare storage technologies using power, energy, duration and efficiency.
- Select a suitable storage mix for a given engineering problem.

Engineering News

As variable solar and wind generation expands, grid operators are deploying batteries, pumped-storage plants and other forms of flexibility. The key challenge is not to find one universal storage device, but to combine technologies operating over seconds, hours, days and seasons.

Main Article

Electricity networks must keep production and consumption balanced at every instant. A sudden mismatch changes grid frequency and can damage equipment or trigger disconnections. Storage systems absorb energy when production exceeds demand and return it when the system needs additional power.

Engineers distinguish **power** from **energy**. Power, measured in watts, describes how quickly a storage system can charge or discharge. Energy, measured in watt-hours or joules, describes how long it can maintain that power. A flywheel may deliver high power for a few seconds, whereas a reservoir or hydrogen cavern can store energy for much longer periods.

Lithium-ion batteries provide rapid response and high round-trip efficiency. They are well suited to frequency control, electric vehicles and daily solar shifting. Their limitations include cost, ageing, fire protection and the supply of raw materials. Pumped-storage hydroelectricity stores much larger amounts of energy: water is pumped to an upper reservoir and later released through a reversible turbine. Its lifetime is long, but suitable terrain and major civil works are required.

Compressed air, thermal storage and hydrogen address longer durations. Hydrogen is produced by electrolysis, stored, and later used in a fuel cell or turbine. Because each conversion loses energy, the round-trip efficiency is lower than that of a battery. Nevertheless, hydrogen may become useful when storage duration matters more than efficiency, particularly for seasonal balancing or industrial fuel supply.

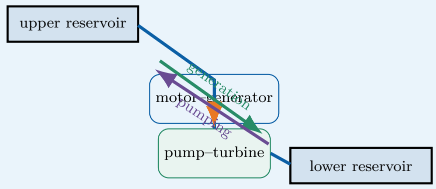
No storage option is optimal for every task. A resilient low-carbon grid therefore combines storage with interconnections, demand response, controllable generation and accurate forecasting.

Did You Know?

A storage system can have excellent efficiency but still be unsuitable if its discharge duration, response time, lifetime or location does not match the service required by the grid.

Engineering Focus

Pumped-storage hydroelectricity

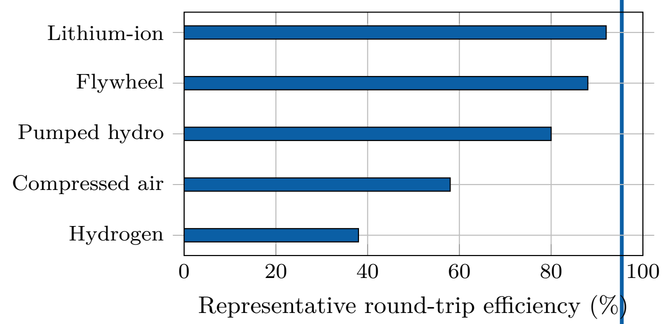


During low demand, electrical energy drives the pump and raises water. During high demand, the water flows downhill and drives the turbine. The stored gravitational energy is $E = mgh$.

Key Figures

Lithium-ion round-trip efficiency	85–95%
Pumped hydro	70–85%
Hydrogen power-to-power	30–45%
Typical battery duration	1–8 h
Seasonal storage	weeks–months

Storage technologies compared



Representative values only: actual performance depends on design and operating conditions.

Useful Vocabulary

round-trip efficiency	rendement aller-retour
charge / discharge	charge / décharge
storage duration	durée de stockage
reservoir	réservoir
flywheel	volant d'inertie
demand response	effacement de consommation
grid frequency	fréquence du réseau

Questions

1. Why must electricity production and consumption remain balanced?
2. Distinguish power from stored energy.
3. Why are batteries suitable for rapid grid services?
4. Explain the operating cycle of a pumped-storage plant.
5. Why can hydrogen be useful despite its lower efficiency?
6. Which criteria should engineers consider besides efficiency?

Engineering Challenge

A remote island has a 12 MW peak demand, a 20 MW wind farm and a 10 MW solar farm. Propose a storage strategy for frequency control, night-time supply and a five-day period of weak wind. Select at least two technologies and justify their power, capacity, duration, efficiency, safety and environmental impact.

Speaking Task

In teams, defend one storage technology in a three-minute technical pitch. Then identify one service for which your technology would be a poor choice.

Sources

IEA – Grid-scale storage: online resource

NREL – Energy storage: online resource

EC06 – Electric Vehicles

Are electric cars really sustainable?

Learning Objectives

- Describe the main components of a battery-electric vehicle powertrain.
- Explain charging, regenerative braking and battery thermal management.
- Assess an electric-vehicle project using technical and environmental criteria.

Engineering News

Electric mobility is moving beyond the vehicle itself. Engineers now design complete systems that connect cars, charging stations, buildings and electricity networks. Smart charging can shift demand away from peak hours, while vehicle-to-grid operation may allow parked vehicles to provide flexibility to the power system.

Did You Know?

An electric motor can reverse its energy flow in a fraction of a second: it drives the wheels during acceleration and becomes a generator during regenerative braking.

Main Article

A battery-electric vehicle stores electrical energy in a high-voltage battery pack. During acceleration, the battery supplies direct current to a power-electronics inverter. The inverter controls the alternating current delivered to the traction motor, which converts electrical energy into mechanical torque at the wheels. Unlike a combustion engine, an electric motor can produce high torque from very low speed and commonly operates with high efficiency over a broad range of conditions.

The battery is not simply a collection of cells. It also includes electrical connections, contactors, sensors, cooling channels, structural protection and a **Battery Management System** (BMS). The BMS estimates state of charge and state of health, balances the cells and protects the pack against excessive current, voltage and temperature. Thermal management is critical because low temperatures reduce available power, while high temperatures accelerate ageing and may increase safety risks.

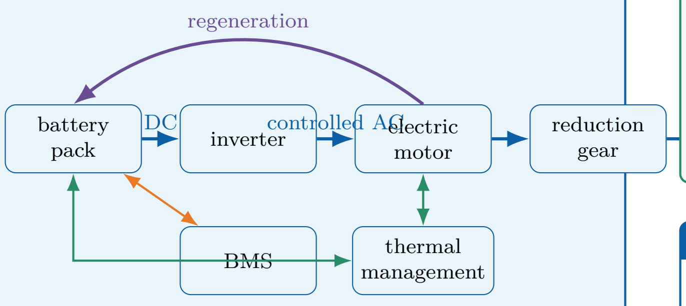
When the driver brakes, the inverter can operate the traction motor as a generator. Part of the vehicle's kinetic energy is then converted back into electrical energy and stored in the battery. This **regenerative braking** reduces energy consumption and mechanical-brake wear, although friction brakes remain necessary for emergency stops, low-speed operation and situations in which the battery cannot accept additional power.

Charging power depends on both the charging station and the vehicle. An on-board charger converts alternating current from a domestic or public AC supply. Fast DC charging bypasses the on-board charger and feeds controlled direct current to the battery. High charging power shortens stops, but it also increases heat generation, infrastructure cost and stress on the local electricity network.

Electric vehicles have no exhaust emissions during use, but they are not impact-free. Engineers must consider battery manufacture, electricity generation, vehicle mass, tyre particles, lifetime and recycling. Their environmental performance therefore depends on the complete life cycle and on how the electricity used for charging is produced.

Engineering Focus

Energy flow in a battery-electric vehicle

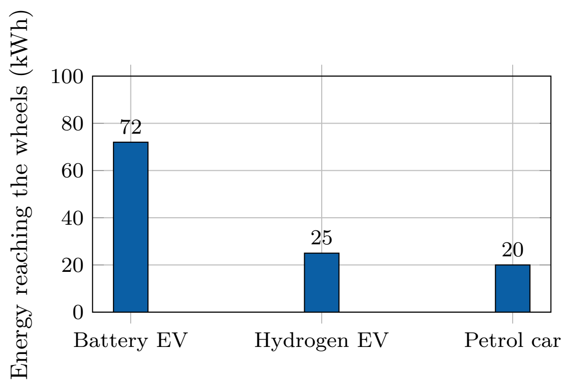


The high-voltage traction circuit is complemented by a DC/DC converter that supplies the vehicle's low-voltage network. Isolation monitoring and contactors disconnect the battery if a serious fault is detected.

Key Figures

Electric motor peak efficiency	90–97%
Typical battery voltage	350–800 V
Home AC charging	2–11 kW
Rapid DC charging	50–350 kW
Typical passenger-car battery	40–100 kWh

From supplied energy to the wheels



Illustrative conversion chain starting from 100 kWh of supplied energy. Actual values vary with technology, driving conditions and system boundaries.

Useful Vocabulary

battery pack	pack batterie
state of charge	état de charge
state of health	état de santé
traction motor	moteur de traction
regenerative braking	freinage régénératif
on-board charger	chargeur embarqué
charging point	point de recharge
power electronics	électronique de puissance

Questions

1. What are the functions of the inverter and the traction motor?
2. Why does a battery pack require thermal management?
3. Explain how regenerative braking changes the direction of energy flow.
4. Compare AC charging with rapid DC charging.
5. Why can charging power be limited by the vehicle rather than the station?
6. Which life-cycle impacts must be considered beyond exhaust emissions?
7. Why is smart charging useful for the electricity network?

Engineering Challenge

A city operates 24 buses, each travelling 220 km per day. Compare overnight depot charging with high-power opportunity charging at selected terminal stops. Propose a battery capacity, charging power and operating schedule. Justify your choices using range, charging time, battery ageing, network connection, redundancy, safety and maintenance.

Speaking Task

Prepare a four-minute technical briefing answering the question: "Should all new urban cars be electric by 2035?" Present one technical advantage, one limitation and one policy measure that could improve the overall transport system.

Sources

IEA – Global EV Outlook: online resource
U.S. Department of Energy – Electric vehicles: online resource
Alternative Fuels Data Center – How electric vehicles work: online resource

EC07 – Robotics

How can robots work safely with humans?

Engineering News

Collaborative robots, often called cobots, are increasingly used in factories, hospitals and laboratories. Unlike traditional industrial robots, they are designed to share a workspace with people.

Main Article

Robots are programmable machines capable of sensing their environment, processing information and performing physical actions. Industrial robots have been used for decades in welding, painting and assembly because they can repeat precise movements at high speed.

A robot generally combines mechanical structures, actuators, sensors, control electronics and software. Electric motors drive the joints, while encoders measure their angular positions. Cameras, force sensors and proximity sensors help the robot understand its surroundings.

Traditional industrial robots normally operate inside safety cages. Collaborative robots are different because they are designed to work near humans. Their control systems monitor speed, force and distance. If a person enters a dangerous area, the robot can slow down or stop automatically.

Robots are also becoming more autonomous. Mobile robots can map buildings, avoid obstacles and transport goods. Medical robots assist surgeons, while agricultural robots can monitor crops and remove weeds.

However, greater autonomy creates new engineering challenges. Robots must remain reliable in unpredictable environments, and their decisions must be understandable. Cybersecurity is also essential because a compromised robot could become dangerous.

Future robots will probably combine better sensors, lighter materials, artificial intelligence and safer control algorithms. Their role will not simply be to replace people, but to perform repetitive, dangerous or highly precise tasks while humans focus on judgement and creativity.

Internet image: collaborative industrial robot

Engineering Focus

Kinematic chain; actuator; encoder; end-effector; trajectory planning; force control; machine vision; obstacle detection; collaborative safety; emergency stop.

Key Figures

Industrial positioning accuracy can reach fractions of a millimetre. Collaborative robots generally operate at lower speeds and forces than conventional industrial robots.

Useful Vocabulary

joint	articulation
actuator	actionneur
encoder	codeur
end-effector	effecteur terminal
workspace	espace de travail
payload	charge utile
trajectory	trajectoire
safety cage	enceinte de sécurité

Questions

1. What are the main components of a robot?
2. Why are encoders necessary?
3. How do collaborative robots improve safety?
4. Give three examples of robot applications.
5. Why is cybersecurity important in robotics?
6. What is the difference between autonomy and automation?
7. Which tasks should remain under human control?

Engineering Challenge

Design a collaborative robotic workstation for assembling a small mechanical product. Specify the sensors, actuators and safety functions required.

Sources

International Federation of Robotics: online resource
NIST Robotics: online resource

EC08 – Artificial Intelligence

Can machines make reliable decisions?

Engineering News

Artificial intelligence is increasingly embedded in engineering systems, from predictive maintenance and autonomous vehicles to medical imaging and industrial quality control.

Main Article

Artificial intelligence refers to computer systems designed to perform tasks that normally require human intelligence. These tasks include recognising images, understanding language, predicting failures and selecting actions.

Many modern AI systems use machine learning. Instead of following only explicit rules, the system learns patterns from examples. A neural network adjusts a large number of internal parameters during training so that its predictions become more accurate.

In engineering, AI can analyse sensor data faster than a human operator. It can detect defects on a production line, estimate the remaining life of a bearing or optimise the energy consumption of a building. AI is also used in autonomous vehicles to identify pedestrians, road signs and other vehicles.

However, AI systems can fail when they encounter situations that were poorly represented in their training data. They may also reproduce biases contained in the data. For safety-critical applications, engineers must therefore test robustness, explain decisions and keep human supervision.

Another challenge is energy consumption. Training large models requires powerful computers and significant electrical energy. Researchers are developing more efficient algorithms and specialised processors.

AI should not be considered an infallible replacement for human judgement. It is a tool that can improve performance when data quality, validation and responsibility are carefully managed.

Internet image: AI used for industrial inspection

Engineering Focus

Training data; neural network; inference; classification; regression; computer vision; predictive maintenance; bias; robustness; explainability.

Key Figures

AI performance strongly depends on the quality and representativeness of training data. A high accuracy score does not guarantee reliable behaviour in every situation.

Useful Vocabulary

dataset	jeu de données
training	entraînement
prediction	prédiction
bias	biais
accuracy	précision
failure	défaillance
pattern	motif
supervision	surveillance humaine

Questions

1. What is the difference between traditional programming and machine learning?
2. What happens during the training of a neural network?
3. Give three engineering applications of AI.
4. Why can an AI system fail in a new situation?
5. What is algorithmic bias?
6. Why is explainability important?
7. Should AI make final decisions in safety-critical systems?

Speaking Task

Should engineers always remain responsible for decisions suggested by artificial intelligence?

Sources

NIST Artificial Intelligence: online resource

OECD Artificial Intelligence: online resource

EC09 – Space Engineering

How do engineers design systems for extreme environments?

Engineering News

Reusable launch vehicles, small satellites and lunar missions are transforming the space sector. Engineers are seeking to reduce launch costs while improving reliability.

Main Article

Spacecraft operate in one of the most demanding environments imaginable. During launch, they experience intense vibration, acceleration and acoustic loads. Once in orbit, they must survive vacuum, radiation and extreme temperature variations.

A satellite includes several subsystems. The structure supports the equipment, solar panels generate electricity and batteries supply power during eclipse. Thermal-control systems prevent components from becoming too hot or too cold. Antennas provide communication with Earth, while onboard computers manage navigation and mission operations.

Attitude control is essential because a satellite must point its instruments, antennas or solar panels in the correct direction. Engineers use reaction wheels, thrusters, gyroscopes and star trackers to measure and control orientation.

Reliability is critical because repairing a satellite is usually impossible. Components are tested under vacuum and vibration before launch, and many functions are duplicated to provide redundancy.

Modern satellites are becoming smaller and more numerous. CubeSats allow universities and small companies to conduct missions at lower cost. At the same time, reusable rockets reduce the price of access to orbit.

Space engineering requires compromises between mass, power, cost and reliability. Every additional kilogram increases launch cost, but insufficient protection may cause mission failure.

Internet image: satellite or reusable launch vehicle

Engineering Focus

Launch loads; vacuum; radiation; thermal control; solar array; battery; telemetry; attitude control; reaction wheel; star tracker; redundancy.

Key Figures

A low-Earth-orbit satellite travels at roughly 7.8 km/s. It may experience repeated transitions between sunlight and darkness during each orbit.

Useful Vocabulary

spacecraft	engin spatial
orbit	orbite
payload	charge utile
thruster	propulseur
vacuum	vide
radiation	rayonnement
attitude	orientation
redundancy	redondance

Questions

1. Which loads affect a spacecraft during launch?
2. Why is thermal control necessary?
3. How does a satellite produce electricity?
4. What is attitude control?
5. Why are systems often duplicated?
6. What are the advantages of CubeSats?
7. Explain the compromise between mass and reliability.

Engineering Challenge

Design the main subsystems of a small Earth-observation satellite. Explain how it will produce power, communicate and control its orientation.

Sources

NASA Small Spacecraft Systems: online resource

European Space Agency: online resource

EC10 – Medical Engineering

How can technology improve healthcare?

Engineering News

Medical devices are becoming more connected, personalised and minimally invasive. Engineers are developing wearable sensors, robotic surgery systems and AI-assisted diagnostics.

Main Article

Medical engineering combines biology, electronics, mechanics, materials science and computer science to improve diagnosis and treatment. Medical devices range from simple thermometers to pacemakers, prosthetic limbs and magnetic resonance imaging systems.

A medical device must be effective, reliable and safe. Sensors measure physical quantities such as temperature, pressure, electrical activity or blood oxygen level. Signal-processing electronics filter noise and convert measurements into useful information for doctors and patients.

Implantable devices create particular challenges. They must use biocompatible materials, consume very little energy and remain reliable for many years. Pacemakers, for example, monitor the heart rhythm and deliver electrical pulses when necessary.

Prosthetic limbs are also becoming more advanced. Some systems detect electrical signals from the user's muscles and use them to control motors. Feedback sensors can help the user estimate grip force or joint position.

Connected medical devices can transmit data remotely, but this creates cybersecurity and privacy risks. A technical failure or malicious attack could affect patient safety. Engineers must therefore protect communication, verify software and comply with strict regulations.

Medical engineering can greatly improve quality of life, but it requires close cooperation between engineers, doctors and patients.

Internet image: prosthetic hand, pacemaker or surgical robot

Engineering Focus

Biomedical sensor; signal conditioning; biocompatibility; implant; prosthesis; closed-loop control; calibration; risk analysis; cybersecurity; clinical validation.

Key Figures

Medical devices may remain implanted for many years and must operate with extremely low failure probabilities. Safety and traceability are central design requirements.

Useful Vocabulary

medical device	dispositif médical
implant	implant
prosthesis	prothèse
heartbeat	battement cardiaque
sensor	capteur
diagnosis	diagnostic
biocompatible	biocompatible
privacy	confidentialité

Questions

1. Which disciplines are combined in medical engineering?
2. Why must sensor signals be filtered?
3. What is the role of a pacemaker?
4. How can a powered prosthesis detect user intention?
5. Why is biocompatibility important?
6. Which cybersecurity risks affect connected medical devices?
7. Why must engineers cooperate with doctors and patients?

Speaking Task

Should connected medical devices automatically transmit patient data to doctors?

Sources

World Health Organization – Medical Devices: online resource

U.S. Food and Drug Administration: online resource

EC11 – Additive Manufacturing

Can 3D printing transform industrial production?

Engineering News

Metal additive manufacturing is increasingly used in aerospace, medical and automotive engineering. It allows engineers to produce lightweight parts with complex internal geometries.

Main Article

Additive manufacturing, commonly known as 3D printing, creates objects by adding material layer by layer. This is different from conventional machining, where material is removed from a larger block.

A digital 3D model is first divided into thin horizontal layers by slicing software. The machine then deposits, melts or solidifies material according to each layer. Different processes can use polymers, metals, ceramics or composite materials.

One major advantage is geometric freedom. Engineers can design internal channels, lattice structures and lightweight shapes that would be difficult or impossible to manufacture with traditional methods. Additive manufacturing is therefore valuable for customised medical implants and aerospace components.

It can also reduce material waste, since only the required quantity of material is deposited. However, the process is not always faster or cheaper. Printing large parts may take many hours, and surfaces often require additional machining.

Mechanical properties can also vary depending on printing direction, temperature and process parameters. Engineers must therefore inspect parts carefully and may use heat treatment to improve performance.

Additive manufacturing is unlikely to replace every conventional process. Its main strength lies in low-volume production, rapid prototyping, customised products and complex high-value components.

Internet image: metal additive manufacturing machine

Engineering Focus

CAD model; slicing; layer thickness; fused deposition modelling; selective laser melting; powder bed; support structure; post-processing; anisotropy; topology optimisation.

Key Figures

Layer thickness may range from a few tens of micrometres to several tenths of a millimetre, depending on the process and material.

Useful Vocabulary

layer	couche
powder	poudre
nozzle	buse
support	support
surface finish	état de surface
prototype	prototype
customised	personnalisé
waste	déchets

Questions

1. What is the difference between additive and subtractive manufacturing?
2. What is the function of slicing software?
3. Why is additive manufacturing suitable for complex shapes?
4. Give two possible materials.
5. Why may post-processing be necessary?
6. What does anisotropy mean in a printed part?
7. Which products are best suited to additive manufacturing?

Engineering Challenge

Redesign a conventional mechanical bracket for additive manufacturing. Reduce mass while preserving stiffness and strength.

Sources

NIST Additive Manufacturing: online resource

NASA Additive Manufacturing: online resource

EC12 – Smart Materials

Can materials react to their environment?

Engineering News

Shape-memory alloys, piezoelectric materials and self-healing polymers are opening new possibilities in aerospace, medicine and civil engineering.

Main Article

Smart materials can change one or more of their properties when they are exposed to temperature, stress, electricity, light or magnetic fields. Unlike passive materials, they can respond actively to their environment.

Shape-memory alloys can recover a predefined shape after deformation when they are heated. Nickel-titanium alloys are used in medical stents, actuators and temperature-controlled mechanisms.

Piezoelectric materials generate an electrical voltage when they are mechanically deformed. The opposite effect is also possible: an applied voltage can produce a small deformation. These materials are used in sensors, ultrasound devices and precision actuators.

Electrochromic materials change colour or transparency when an electrical voltage is applied. Smart windows can therefore reduce the amount of solar radiation entering a building and lower cooling energy consumption.

Researchers are also developing self-healing materials. Some polymers contain microcapsules filled with repair agents. When a crack forms, the capsules break and release the agent, partially restoring the material.

These technologies can reduce maintenance and improve system performance, but they may be expensive and difficult to recycle. Engineers must also understand how their properties change over time and under repeated loading.

Internet image: shape-memory alloy or smart window

Engineering Focus

Shape-memory effect; phase transformation; piezoelectricity; electrochromism; self-healing polymer; smart sensor; actuator; fatigue; ageing.

Useful Vocabulary

alloy	alliage
deformation	déformation
crack	fissure
healing	réparation
voltage	tension
strain	déformation relative
stiffness	rigidité
lifetime	durée de vie

Questions

1. What makes a material “smart”?
2. How does a shape-memory alloy recover its shape?
3. Explain the direct piezoelectric effect.
4. Give two applications of piezoelectric materials.
5. How can smart windows save energy?
6. How do self-healing polymers work?
7. Which limitations could slow the adoption of smart materials?

Speaking Task

Which smart material could have the greatest impact on everyday life?

Sources

European Space Agency – Smart Materials: online resource

NIST Materials Research: online resource

EC13 – Smart Cities

Can connected systems make cities more sustainable?

Engineering News

Cities are deploying connected sensors to manage traffic, lighting, waste collection and energy consumption in real time.

Main Article

A smart city uses digital technologies and connected infrastructure to improve services, reduce resource consumption and make urban life more efficient.

Sensors can measure traffic flow, air quality, noise, temperature and electricity use. These data are transmitted through communication networks and analysed by software platforms.

Smart traffic lights can adapt their timing to vehicle flow. Connected parking systems can guide drivers directly to available spaces, reducing congestion and fuel consumption. Street lighting can automatically dim when no pedestrians or vehicles are present.

Buildings are another major area of improvement. Automated systems can adjust heating, ventilation and lighting according to occupancy and weather conditions. Smart grids can coordinate local renewable generation, batteries and electric-vehicle charging.

However, collecting large quantities of data creates privacy and cybersecurity concerns. A connected city must protect personal information and keep essential services operational during technical failures or cyberattacks.

Technology alone cannot solve every urban problem. Smart-city projects must also consider accessibility, cost, public acceptance and social inequalities.

Internet image: connected urban infrastructure

Engineering Focus

Internet of Things; sensor network; data platform; smart grid; adaptive traffic control; digital twin; interoperability; cybersecurity; privacy.

Key Figures

Urban systems may generate data continuously from thousands of sensors. Reliable communication and energy-efficient devices are therefore essential.

Useful Vocabulary

traffic flow	flux de circulation
street lighting	éclairage public
waste collection	collecte des déchets
occupancy	occupation
privacy	vie privée
outage	panne
urban planning	urbanisme
real time	temps réel

Questions

1. What is the objective of a smart city?
2. Which quantities can urban sensors measure?
3. How can adaptive traffic lights reduce congestion?
4. How can smart buildings save energy?
5. Why is interoperability important?
6. Which privacy risks exist?
7. Can a city be smart without being highly connected?

Engineering Challenge

Design a smart system for reducing energy consumption in a school district. Specify sensors, communication and control strategies.

Sources

European Commission – Smart Cities: online resource

NIST Smart Cities: online resource

EC14 – Water Treatment

How can engineering provide safe drinking water?

Engineering News

Water-treatment technologies such as membrane filtration, ultraviolet disinfection and solar desalination are being developed to improve access to safe drinking water.

Main Article

Safe drinking water requires the removal of particles, microorganisms and chemical pollutants. A treatment plant usually combines several physical and chemical processes.

Large particles are first removed by screening. Coagulation causes smaller suspended particles to join together, forming larger flocs. These flocs settle by gravity during sedimentation.

The water then passes through filters made of sand, activated carbon or membranes. Filtration removes remaining particles and some dissolved substances. Disinfection using chlorine, ozone or ultraviolet light destroys harmful microorganisms.

In coastal or arid regions, desalination can produce freshwater from seawater. Reverse osmosis uses pressure to force water through a membrane that blocks salts. This process is effective but consumes significant energy.

Engineers must also monitor water quality continuously. Sensors measure properties such as turbidity, pH, conductivity and chlorine concentration.

Water treatment is not only a technical issue. Systems must be affordable, maintainable and adapted to local conditions. In remote communities, simple and robust solutions may be more useful than highly complex equipment.

Internet image: water treatment or reverse-osmosis plant

Engineering Focus

Screening; coagulation; flocculation; sedimentation; filtration; activated carbon; disinfection; reverse osmosis; turbidity sensor; water-quality monitoring.

Useful Vocabulary

drinking water	eau potable
pollutant	polluant
membrane	membrane
pressure	pression
salt	sel
microorganism	micro-organisme
turbidity	turbidité
treatment plant	usine de traitement

Questions

1. Why are several treatment stages necessary?
2. What is the purpose of coagulation?
3. How does sedimentation work?
4. Compare filtration and disinfection.
5. Explain reverse osmosis.
6. Why does desalination require significant energy?
7. Which solution would be suitable for a remote village?

Engineering Challenge

Design a compact water-treatment system for a village located near a polluted river. Consider reliability, maintenance and energy use.

Sources

World Health Organization – Drinking Water: online resource

U.S. Environmental Protection Agency: online resource

EC15 – Future Transport

How will people and goods move tomorrow?

Engineering News

Autonomous shuttles, electric aircraft, hydrogen trains and high-speed rail are being developed to reduce the environmental impact of transport.

Main Article

Transport systems must move people and goods safely, quickly and efficiently. However, they are also responsible for significant energy consumption and greenhouse-gas emissions.

Future transport will probably combine several technologies rather than rely on a single solution. Battery-electric vehicles are suitable for many short and medium journeys. Hydrogen may be useful for heavy trucks, ships or trains when batteries would be too large or heavy.

Rail transport can move large numbers of passengers with relatively low energy consumption. High-speed trains can replace some short-haul flights, especially when city centres are connected directly.

Autonomous vehicles use cameras, radar, lidar and satellite navigation to perceive their environment. Their onboard computers must identify obstacles, predict the behaviour of other road users and select safe trajectories.

Urban mobility is also changing. Shared bicycles, electric buses and on-demand shuttles can reduce congestion if they are integrated into an efficient public-transport network.

The main challenge is not only technical. Engineers and planners must consider infrastructure, cost, accessibility and public acceptance. A highly advanced system is not sustainable if only a small part of the population can use it.

Internet image: autonomous shuttle or high-speed train

Engineering Focus

Battery-electric propulsion; hydrogen fuel cell; high-speed rail; autonomous navigation; radar; lidar; trajectory planning; mobility as a service; multimodal transport.

Useful Vocabulary

journey	trajet
freight	fret
commuter	usager pendulaire
shared mobility	mobilité partagée
short-haul flight	vol court-courrier
road user	usager de la route
congestion	embouteillage
accessibility	accessibilité

Questions

1. Why will future transport combine several technologies?
2. When may hydrogen be preferable to batteries?
3. Why can high-speed rail replace some flights?
4. Which sensors are used by autonomous vehicles?
5. What is trajectory planning?
6. How can shared mobility reduce congestion?
7. Which criteria define a sustainable transport system?

Speaking Task

Should city centres ban private cars and rely mainly on public and shared transport?

Sources

International Energy Agency – Transport: online resource

European Commission – Mobility and Transport: online resource

EC16 – Cybersecurity

How can engineers protect connected systems?

Engineering News

Current developments illustrate the rapid evolution of this engineering field and its impact on society.

Main Article

Cybersecurity is an important domain of modern engineering. Engineers combine scientific knowledge, modelling, experimentation and digital technologies to design systems that are safer, more efficient and more sustainable.

A complete engineering system integrates sensors, actuators, electronics, software and communication. Design choices always involve compromises between performance, cost, reliability, safety and environmental impact.

Engineers analyse requirements, model the system, build prototypes, test performance and continuously improve the final solution. International standards and risk analysis play a central role throughout the design process.

Future innovations will rely on multidisciplinary teams capable of combining mechanics, electronics, computer science, materials engineering and artificial intelligence.

Internet illustration for Cybersecurity

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; reliability; sustainability; innovation.

Useful Vocabulary

system	système
sensor	capteur
actuator	actionneur
software	logiciel
reliability	fiabilité
safety	sécurité
maintenance	maintenance
innovation	innovation

Questions

1. Explain the engineering problem addressed in this chapter.
2. Which disciplines are involved?
3. Why is modelling useful?
4. Why is testing essential?
5. What are the main technical challenges?
6. Which environmental impacts should engineers consider?
7. Which future innovation seems the most promising?

Engineering Challenge

Propose an engineering solution related to this topic and justify your technical choices.

Sources

IEEE: online resource

EC17 – Internet of Things

Connecting billions of devices

Engineering News

Current developments illustrate the rapid evolution of this engineering field and its impact on society.

Main Article

Internet of Things is an important domain of modern engineering. Engineers combine scientific knowledge, modelling, experimentation and digital technologies to design systems that are safer, more efficient and more sustainable.

A complete engineering system integrates sensors, actuators, electronics, software and communication. Design choices always involve compromises between performance, cost, reliability, safety and environmental impact.

Engineers analyse requirements, model the system, build prototypes, test performance and continuously improve the final solution. International standards and risk analysis play a central role throughout the design process.

Future innovations will rely on multidisciplinary teams capable of combining mechanics, electronics, computer science, materials engineering and artificial intelligence.

Internet illustration for Internet of Things

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; reliability; sustainability; innovation.

Useful Vocabulary

system	système
sensor	capteur
actuator	actionneur
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Engineering Challenge

Propose an engineering solution related to this topic and justify your technical choices.

Sources

IEEE: online resource

EC18 – Drones

Engineering autonomous aerial systems

Engineering News

Current developments illustrate the rapid evolution of this engineering field and its impact on society.

Main Article

Drones is an important domain of modern engineering. Engineers combine scientific knowledge, modelling, experimentation and digital technologies to design systems that are safer, more efficient and more sustainable.
A complete engineering system integrates sensors, actuators, electronics, software and communication. Design choices always involve compromises between performance, cost, reliability, safety and environmental impact.
Engineers analyse requirements, model the system, build prototypes, test performance and continuously improve the final solution. International standards and risk analysis play a central role throughout the design process.
Future innovations will rely on multidisciplinary teams capable of combining mechanics, electronics, computer science, materials engineering and artificial intelligence.

Internet illustration for Drones

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; reliability; sustainability; innovation.

Useful Vocabulary

system	système
sensor	capteur
actuator	actionneur
software	logiciel
reliability	fiabilité
safety	sécurité
maintenance	maintenance
innovation	innovation

Questions

1. Explain the engineering problem addressed in this chapter.
2. Which disciplines are involved?
3. Why is modelling useful?
4. Why is testing essential?
5. What are the main technical challenges?
6. Which environmental impacts should engineers consider?
7. Which future innovation seems the most promising?

Engineering Challenge

Propose an engineering solution related to this topic and justify your technical choices.

Sources

IEEE: online resource

EC19 – Biomimetics

Learning from nature

Engineering News

Current developments illustrate the rapid evolution of this engineering field and its impact on society.

Main Article

Biomimetics is an important domain of modern engineering. Engineers combine scientific knowledge, modelling, experimentation and digital technologies to design systems that are safer, more efficient and more sustainable.

A complete engineering system integrates sensors, actuators, electronics, software and communication. Design choices always involve compromises between performance, cost, reliability, safety and environmental impact.

Engineers analyse requirements, model the system, build prototypes, test performance and continuously improve the final solution. International standards and risk analysis play a central role throughout the design process.

Future innovations will rely on multidisciplinary teams capable of combining mechanics, electronics, computer science, materials engineering and artificial intelligence.

Internet illustration for Biomimetics

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; reliability; sustainability; innovation.

Useful Vocabulary

system	système
sensor	capteur
actuator	actionneur
software	logiciel
reliability	fiabilité
safety	sécurité
maintenance	maintenance
innovation	innovation

Questions

1. Explain the engineering problem addressed in this chapter.
2. Which disciplines are involved?
3. Why is modelling useful?
4. Why is testing essential?
5. What are the main technical challenges?
6. Which environmental impacts should engineers consider?
7. Which future innovation seems the most promising?

Engineering Challenge

Propose an engineering solution related to this topic and justify your technical choices.

Sources

IEEE: online resource

EC20 – Circular Economy

Designing products for sustainability

Engineering News

Current developments illustrate the rapid evolution of this engineering field and its impact on society.

Main Article

Circular Economy is an important domain of modern engineering. Engineers combine scientific knowledge, modelling, experimentation and digital technologies to design systems that are safer, more efficient and more sustainable.

A complete engineering system integrates sensors, actuators, electronics, software and communication. Design choices always involve compromises between performance, cost, reliability, safety and environmental impact.

Engineers analyse requirements, model the system, build prototypes, test performance and continuously improve the final solution. International standards and risk analysis play a central role throughout the design process.

Future innovations will rely on multidisciplinary teams capable of combining mechanics, electronics, computer science, materials engineering and artificial intelligence.

Internet illustration for Circular Economy

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; reliability; sustainability; innovation.

Useful Vocabulary

system	système
sensor	capteur
actuator	actionneur
software	logiciel
reliability	fiabilité
safety	sécurité
maintenance	maintenance
innovation	innovation

Questions

1. Explain the engineering problem addressed in this chapter.
2. Which disciplines are involved?
3. Why is modelling useful?
4. Why is testing essential?
5. What are the main technical challenges?
6. Which environmental impacts should engineers consider?
7. Which future innovation seems the most promising?

Engineering Challenge

Propose an engineering solution related to this topic and justify your technical choices.

Sources

IEEE: online resource

EC21 – Digital Twin

Creating virtual replicas of physical systems

Engineering News

Recent advances illustrate how Digital Twin is transforming engineering.

Main Article

Digital Twin combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.

Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.

Future developments will rely on multidisciplinary engineering teams capable of designing sustainable and resilient technical systems.

Internet illustration: Digital Twin

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; sustainability; reliability; innovation.

Useful Vocabulary

system	système
model	modèle
sensor	capteur
simulation	simulation
performance	performances
reliability	fiabilité
efficiency	rendement
innovation	innovation

Questions

1. What engineering problem is addressed?
2. Which scientific disciplines are involved?
3. Why is modelling important?
4. Which constraints must engineers consider?
5. Which innovation seems the most promising?

Engineering Challenge

Propose a realistic engineering solution related to this topic.

Sources

IEEE: online resource

EC22 – Industry 4.0

Connected manufacturing

Engineering News

Recent advances illustrate how Industry 4.0 is transforming engineering.

Main Article

Industry 4.0 combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.

Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.

Future developments will rely on multidisciplinary engineering teams capable of designing sustainable and resilient technical systems.

Internet illustration: Industry 4.0

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; sustainability; reliability; innovation.

Useful Vocabulary

system	système
model	modèle
sensor	capteur
simulation	simulation
performance	performances
reliability	fiabilité
efficiency	rendement
innovation	innovation

Questions

1. What engineering problem is addressed?
2. Which scientific disciplines are involved?
3. Why is modelling important?
4. Which constraints must engineers consider?
5. Which innovation seems the most promising?

Engineering Challenge

Propose a realistic engineering solution related to this topic.

Sources

IEEE: online resource

EC23 – Autonomous Vehicles

Engineering safe autonomous mobility

Engineering News

Recent advances illustrate how Autonomous Vehicles is transforming engineering.

Main Article

Autonomous Vehicles combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.

Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.

Future developments will rely on multidisciplinary engineering teams capable of designing sustainable and resilient technical systems.

Internet illustration: Autonomous Vehicles

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; sustainability; reliability; innovation.

Useful Vocabulary

system	système
model	modèle
sensor	capteur
simulation	simulation
performance	performances
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Questions

1. What engineering problem is addressed?
2. Which scientific disciplines are involved?
3. Why is modelling important?
4. Which constraints must engineers consider?
5. Which innovation seems the most promising?

Engineering Challenge

Propose a realistic engineering solution related to this topic.

Sources

IEEE: online resource

EC24 – Smart Grids

Managing tomorrow's electricity networks

Engineering News

Recent advances illustrate how Smart Grids is transforming engineering.

Main Article

Smart Grids combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.

Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.

Future developments will rely on multidisciplinary engineering teams capable of designing sustainable and resilient technical systems.

Internet illustration: Smart Grids

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; sustainability; reliability; innovation.

Useful Vocabulary

system	système
model	modèle
sensor	capteur
simulation	simulation
performance	performances
reliability	fiabilité
efficiency	rendement
innovation	innovation

Questions

1. What engineering problem is addressed?
2. Which scientific disciplines are involved?
3. Why is modelling important?
4. Which constraints must engineers consider?
5. Which innovation seems the most promising?

Engineering Challenge

Propose a realistic engineering solution related to this topic.

Sources

IEEE: online resource

EC25 – Marine Renewable Energy

Power from oceans

Engineering News

Recent advances illustrate how Marine Renewable Energy is transforming engineering.

Main Article

Marine Renewable Energy combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.

Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.

Future developments will rely on multidisciplinary engineering teams capable of designing sustainable and resilient technical systems.

Internet illustration: Marine Renewable Energy

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; sustainability; reliability; innovation.

Useful Vocabulary

system	système
model	modèle
sensor	capteur
simulation	simulation
performance	performances
reliability	fiabilité
efficiency	rendement
innovation	innovation

Questions

1. What engineering problem is addressed?
2. Which scientific disciplines are involved?
3. Why is modelling important?
4. Which constraints must engineers consider?
5. Which innovation seems the most promising?

Engineering Challenge

Propose a realistic engineering solution related to this topic.

Sources

IEEE: online resource

EC26 – Biomedical Imaging

Seeing inside the human body

Engineering News

Recent advances illustrate how Biomedical Imaging is transforming engineering.

Main Article

Biomedical Imaging combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.

Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.

Future developments will rely on multidisciplinary engineering teams capable of designing sustainable and resilient technical systems.

Internet illustration: Biomedical Imaging

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; sustainability; reliability; innovation.

Useful Vocabulary

system	système
model	modèle
sensor	capteur
simulation	simulation
performance	performances
reliability	fiabilité
efficiency	rendement
innovation	innovation

Questions

1. What engineering problem is addressed?
2. Which scientific disciplines are involved?
3. Why is modelling important?
4. Which constraints must engineers consider?
5. Which innovation seems the most promising?

Engineering Challenge

Propose a realistic engineering solution related to this topic.

Sources

IEEE: online resource

EC27 – Nanotechnology

Engineering at the nanoscale

Engineering News

Recent advances illustrate how Nanotechnology is transforming engineering.

Main Article

Nanotechnology combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.

Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.

Future developments will rely on multidisciplinary engineering teams capable of designing sustainable and resilient technical systems.

Internet illustration: Nanotechnology

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; sustainability; reliability; innovation.

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IEEE: online resource

EC28 – Quantum Technologies

The next computing revolution

Engineering News

Recent advances illustrate how Quantum Technologies is transforming engineering.

Main Article

Quantum Technologies combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.

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Internet illustration: Quantum Technologies

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IEEE: online resource

EC29 – Sustainable Buildings

Designing low-carbon buildings

Engineering News

Recent advances illustrate how Sustainable Buildings is transforming engineering.

Main Article

Sustainable Buildings combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.

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Internet illustration: Sustainable Buildings

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IEEE: online resource

EC30 – High-Speed Transportation

Moving faster with less energy

Engineering News

Recent advances illustrate how High-Speed Transportation is transforming engineering.

Main Article

High-Speed Transportation combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.

Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.

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Internet illustration: High-Speed Transportation

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